BURHAN ARSHAD

AI & Machine Learning Developer | Computer Science Student

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PROFILE

Motivated Computer Science student at the University of Central Punjab with a CGPA of 3.99/4.00 and a strong focus on machine learning, deep learning, natural language processing and computer vision. Works across the full lifecycle of intelligent systems: data processing, model development, evaluation, API design, deployment and integration into real applications. Experienced in retrieval-augmented generation, CNN and LSTM modeling, and FastAPI-based inference services, with additional freelance experience in WordPress and Shopify development and a hobby background in Unity game development. Guided by the philosophy: Learn. Build. Break. Improve. Deploy.

EDUCATION

Bachelor of Science in Computer Science — Pakistan

University of Central Punjab (UCP) | Currently in 4th Semester

- Cumulative GPA: **3.99 / 4.00**
- Focus areas: Machine Learning, Deep Learning, NLP, Computer Vision, Generative AI, Agentic AI.

TECHNICAL SKILLS

Programming	Python, C#, C++, JavaScript, SQL, HTML, CSS
Machine Learning	Scikit-learn, TensorFlow, Keras, Random Forest, Linear SVM, GridSearchCV, NumPy, Pandas
Deep Learning & CV	CNNs, LSTM, Image Classification, Behavioral Cloning, OpenCV
NLP & Generative AI	TF-IDF, Text Classification, LangChain, ChromaDB, Sentence-Transformers, Hugging Face, Groq, Retrieval-Augmented Generation (RAG), Agentic AI
Data & Visualization	NumPy, Pandas, Matplotlib, Seaborn, Jupyter Notebook
Backend & APIs	FastAPI, Uvicorn, REST APIs, WebSockets / Realtime Systems
Databases	PostgreSQL, Supabase, SQL
Frontend & Web	Astro, HTML, CSS, JavaScript, Tailwind CSS, GSAP, Framer Motion, WordPress, Shopify (Liquid)
Deployment	Streamlit Cloud, Vercel, Render
Game Development	Unity, C#, Blender, Netcode for GameObjects, ParrelSync
Tools	Git, GitHub, VS Code, Virtual Environments

MACHINE LEARNING & AI PROJECTS

Real-Time ASL Sign Language Recognition — Live Demo • GitHub

Python, TensorFlow/Keras, CNN, Computer Vision, FastAPI, WebSockets

- Designed and trained a custom CNN on 87,000 images across 29 classes to recognize American Sign Language alphabet gestures, reaching 99.19% validation accuracy.
- Built a FastAPI WebSocket pipeline that returns real-time predictions from a live webcam feed; deployed on Render.

Self-Driving Car: Behavioral Cloning — GitHub

Python, TensorFlow/Keras, CNN, Computer Vision

- Implemented an end-to-end driving model inspired by NVIDIA's PilotNet architecture that maps camera frames directly to steering angles.
- Applied multi-camera training, data augmentation and steering-distribution balancing, with real-time simulator control.

RAG Document Assistant Live Demo • GitHub

Python, LangChain, ChromaDB, Sentence-Transformers, Groq, Streamlit

- Developed a Retrieval-Augmented Generation application that answers questions about uploaded PDF or TXT documents.
- Chunked and embedded documents locally, stored vectors in ChromaDB, retrieved context with MMR and generated responses through a Groq-hosted LLM.
- Displayed retrieved source chunks alongside answers to improve traceability.

CineMatch: Movie Recommendation System Live Demo • GitHub

Python, Scikit-learn, TF-IDF, Cosine Similarity, FastAPI, Streamlit, TMDB API

- Built a content-based recommendation engine using TF-IDF vectors and cosine similarity.
- Separated a FastAPI backend from a Streamlit frontend and integrated live TMDB data for search, posters, ratings and genre-based recommendations.

Chest X-Ray Pneumonia Detection Live Demo • GitHub

Python, TensorFlow/Keras, CNN, Computer Vision, Streamlit

- Developed a medical-imaging CNN that distinguishes normal chest X-rays from pneumonia cases.
- Prioritized pneumonia-class recall (94%) to reduce missed positive cases; deployed as an interactive Streamlit app.

BTC Directional Signal: LSTM with Backtested Evaluation Live Demo • GitHub

Python, TensorFlow/Keras, LSTM, Pandas, Streamlit

- Trained a time-series LSTM to predict short-term Bitcoin price direction from 100,000 hourly candles and 15 engineered technical indicators.
- Evaluated through out-of-sample backtesting covering transaction costs, Sharpe ratio, maximum drawdown and win rate rather than classification accuracy alone.

SMS Spam Classifier Live Demo • GitHub

Python, Scikit-learn, TF-IDF, LinearSVC, GridSearchCV, Streamlit

- Classified SMS messages as spam or legitimate using TF-IDF features and a Linear SVM tuned with GridSearchCV, achieving ~99% accuracy and ~97% macro F1-score.
- Deployed a real-time prediction interface with Streamlit.

Bike Demand Prediction Live Demo • GitHub

Python, Pandas, Scikit-learn, Random Forest, GridSearchCV, Streamlit

- Built a regression model that predicts bike rental demand from environmental and temporal features using a tuned Random Forest.

SOFTWARE ENGINEERING & CLOUD PROJECTS

Cloud-Based Multi-Tenant SaaS Inventory System Live Demo • GitHub

PostgreSQL, Supabase, SQL, TypeScript, Realtime/WebSockets

- Designed a multi-tenant inventory management platform with strict tenant isolation, relational database architecture and authentication.
- Implemented real-time data synchronization and deployed the application on Vercel.

GAME DEVELOPMENT PROJECTS

Unity Multiplayer Stealth-Comedy Burglary Game GitHub

Unity, C#, Netcode for GameObjects, ParrelSync, Blender

- Developed a multiplayer game featuring networked player movement, interactive environments, enemy AI behavior, and cooperative and competitive mechanics.

Debug or Die GitHub

Unity, C#, Blender

- Created a narrative-driven horror survival game focused on atmosphere, pacing, environmental interaction and interactive storytelling.

Two-Player Console Chess GitHub

C++

- Implemented complete two-player chess game logic from scratch as a console application.

PROFESSIONAL EXPERIENCE

Freelance Web Developer Remote

Upwork & Independent Clients

- Developed a professional WordPress website for Pearl Essence Interiors, a UAE-based interior design and fit-out business, showcasing its services, projects, brand identity and consultation experience.
- Built a Shopify eCommerce store for KADME, a Pakistani footwear brand, focusing on product presentation, collection organization, responsive layouts and conversion-focused design.
- Offer services spanning machine learning solutions, NLP and generative AI applications, data analysis and visualization, custom software and eCommerce development.

LEADERSHIP & ACTIVITIES

Coding Club Coordinator

University of Central Punjab

- Coordinate coding-related sessions and presentations for the university Coding Club.

ONLINE PRESENCE

Portfolio: burhan.cypherai.tech
GitHub: github.com/burhan-arshad
LinkedIn: linkedin.com/in/burhan-arshad
Upwork: Freelancer Profile

REFERENCES

Available upon request.